Walchand Institute of Technology, Solapur
(An Autonomous Institute)

Affiliated to
Punyashlok Ahilyadevi Holkar Solapur University,
Solapur

Choice Based Credit System (CBCS)
Structure and Syllabus
Multidisciplinary Minor Program in Electronics and Computer Engineering
offered by
Department of Electronics Engineering
for Batch Admitted in
2023-24

Dean(Academics)

Walchand Institute of Technology Electronics Dept. Solapur.

HEAD
Electronics Department
Walchand Institute of Technology
SOLAPUR-413006.

Multidisciplinary Minor Program in Electronics and Computer Engineering	
Legends Used	
L	Lecture Hours / week
T	Tutorial Hours / week
P	Practical Hours / week
FA	Formative Assessment
SA	Summative Assessment
ESE	End Semester Examination
ISE	In Semester Evaluation
ICA	Internal Continuous Assessment
POE	Practical and Oral Exam
OE	Oral Exam
MOOC	Massive Open Online Course
HSS	Humanity and Social Science
NPTEL	National Programme on Technology Enhanced Learning
F.Y.	First Year
S.Y.	Second Year
T.Y.	Third Year
B.Tech.	Bachelor of Technology

Multidisciplinary Minor Program in Electronics and Computer Engineering

Course Code Format

2	4	E	C	U/P	2	C	C	1	T/L
Year of syllabus revision		Program Code		U-Under Graduate, P-Post Graduate	Semester No./ Year1/2/3/…8	Course Type		Course Serial No. 1-9	T-Theory, L-Lab session A-Tutorial P-Programming / Design

Program Code	
ET	Electronics and Telecommunication Engineering
Course Type	
BS	Basic Science
ES	Engineering Science
HU	Humanities & Social Science
MC	Mandatory Course
CC	Programme Core Compulsory Course
SN*	Self-Learning (**N*** *indicates the serial number of electives offered in the respective category)*
EN*	Programme Core Elective Course (**N*** *indicates the serial number of electives offered in the respective category)*
SK	Skill-Based Course
SM	Seminar
MP	Mini project
PR	Project
IN	Internship
ON*	Open Elective (**N*** *indicates the serial number of electives offered in the respective category)*
MD	Multidisciplinary Minor

EM	Entrepreneurship/Economics/Management Courses
FP	Community Engagement Project / Field Project
AE	Ability Enhancement Course
VE	Value Education Course
IK	Indian Knowledge System
VS	Vocational & Skill Enhancement Course
RM	Research Methodology
HN	Honors' Degree Course
HR	Honors Research

Sample Course Code	
23ECU3MD6T	Analog Electronics

Multidisciplinary Minor Program in Electronics and Computer Engineering

Semester	Course Code	Name of Course	Engagement Hours			Credits					Total
			L	T	P		Theory	OE/ POE	ISE	ICA	
III	23ECU3MD6T	Analog Electronics	2	-	-	2	60	-	40	-	100
IV	23ECU4MD6T	Digital Electronics	1	-	-	1		-	50	-	50
V	23ECU5MD6T	Microcontrollers and Peripherals	3	-	-	3	60	-	40	-	100
VI	23ECU6MD6T	Advanced Controllers and Interfacing	2	-	-	2	60	-	40	-	100
VII	23ECU7MD6T	Electronic System Design	3	-	-	3	60	-	40	-	100
VII	23ECU7MD6A	Electronic System Design(Tutorial)	-	1	-	1	-	-	-	25	25
		Sub-Total	**11**	**1**	**0**	**12**	**240**		**210**	**25**	**475**
Laboratory Courses											
IV	23ECU4MD6L	Digital Electronics	-	-	2	1	-	-	-	25	25
VI	23ECU6MD6L	Advanced Controllers and Interfacing	-	-	2	1	-	-	-	25	25
		Sub-Total	**0**	**0**	**4**	**2**	**0**		**0**	**50**	**50**
		Grand-Total	**11**	**1**	**4**	**14**	**240**		**210**	**75**	**525**

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23ECU7MDM6T: Electronic System Design

Teaching Scheme		Examination Scheme	
Lectures	3 Hours/week	**ESE**	60 Marks
Tutorial	1 Hour/week	**ISE**	40 Marks
Credits	4	**ICA**	25 Marks

Introduction:

Electronic System Design focuses on the complete process of developing an electronic product, starting from power supply design and signal conditioning to controller selection, PCB realization, and professional documentation. The course emphasizes system-level thinking, practical design considerations, and prototype development using modern microcontrollers and design tools.

Course Prerequisite:

Students should have prior knowledge of Basic Electronics, Digital Electronics, Microcontroller 8051 and Peripherals, ESP32 and Interfacing.

Course Objectives:

1. To make students aware about the design process of electronic systems from concept to implementation.
2. To design power supply and signal conditioning circuits for embedded systems.
3. To select suitable controllers and hardware components based on system requirements.
4. To develop PCB layouts and professional electronics project documentation.

Course Outcomes:

After completing this course, student will be able to -

1. Design power supply circuits required for embedded electronic systems.
2. Develop sensor–controller–actuator based electronic prototypes.
3. Design basic PCBs for electronic products using open-source tools.
4. Prepare complete engineering documentation for an electronic system.

Unit – I	**Electronics Power Supply Design**	**08 Hours**

Role of power supply in electronic systems, AC and DC power sources, rectifiers, Filters, Voltage regulators, Battery technologies used in embedded systems, Battery Management Systems (BMS) fundamentals, Power design considerations for embedded devices and systems.

Unit – II	**Signal Conditioning Circuits**	**08 Hours**

Need for signal conditioning in electronic systems, Sensor output characteristics, Amplification using operational amplifiers, Filtering and noise reduction techniques, Analog to digital signal conditioning,

Driver circuits for actuators (relay, transistor, drivers), Protection circuits for sensor and actuator interfaces.		
Unit – III	**System Requirements and Controller Selection**	**08 Hours**
Role of controllers in electronic systems, Criteria for controller selection, Processing power, Memory requirements, Communication interfaces, Power consumption, Cost considerations, Comparison of controllers, Peripheral integration requirements, Case study: selecting a controller for an IoT device		
Unit – IV	**PCB Design Considerations**	**08 Hours**
Introduction to PCB design workflow, Schematic capture basics, Component footprints and libraries, Routing techniques and ground planes, Design rule check (DRC), Introduction to open-source PCB tools (KiCad / EasyEDA).		
Unit – V	**Electronics Project Documentation**	**08 Hours**
Importance of engineering documentation, Project proposal and requirement specification, System block diagram and architecture, Circuit diagram and design explanation, PCB design documentation, Firmware flowchart and program explanation, Bill of Materials (BOM) preparation, Testing procedures and results documentation, User manual preparation.		

Internal Continuous Assessment (ICA)

ICA shall consist of minimum Six Tutorials. Also, students shall complete a mini-project in group applying their acquired skills and knowledge during this course.

1. Block Diagram Development for an Electronic System
2. Power Supply Design for Embedded Systems
3. Battery Selection and BMS Configuration
4. Sensor Signal Conditioning Circuit Design
5. Actuator Driver Circuit Design
6. Controller Selection for Embedded Applications
7. PCB Layout Planning and Component Placement
8. Electronics Project Documentation Preparation

Text Books

1. Raj Kamal, Embedded Systems: Architecture, Programming and Design, McGraw Hill Education.
2. Paul Scherz and Simon Monk, Practical Electronics for Inventors, McGraw Hill.
3. Muhammad Ali Mazidi, Janice Gillispie Mazidi, and Rolin McKinlay, The 8051 Microcontroller and Embedded Systems, Pearson Education.
4. Simon Monk, Programming ESP32: Getting Started with the ESP32 Using Arduino IDE, McGraw Hill.

Reference Books
1. Bruce R. Archambeault and James Drewniak, PCB Design for Real-World EMI Control, Springer. 2. John Catsoulis, Designing Embedded Hardware, O'Reilly Media. 3. Horowitz and Hill, The Art of Electronics, Cambridge University Press. 4. R.G. Kaduskar and V.B.Baru, Electronic Product design, Wiley India.